

Summable covariance jets and a growing affine-root-union uniform neighbourhood

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-08-03 · this version issued 2026-08-03 · v1.0

1. Purpose and scope: the cardinality-uniform neighbourhood gap

R-160 proves a strict origin-Hessian gap for every finite R-155 family of one-shot fixed-law affine controls, uniformly in the number and lengths of the factor-three components. Its open boundary was whether one radius could be chosen independently of the family cardinality. Finite-dimensional continuity for each family does not answer that question.

Let $\mathcal{P}$ be any finite set of distinct nonzero side-16 dual-lattice momentum classes modulo sign. Retain $[2p]$ for every $p \in \mathcal{P}$ and allow $\mathcal{P} \cap 2\mathcal{P} \neq \emptyset$; the finite cutoff retains the whole union $\mathcal{P} \cup 2\mathcal{P}$. On the common fixed-law Gaussian space, with $A_p \in \mathbb{R}^{12 \times 12}$, set

$$h_{[2p]} = A_p \xi_{[p]}, \quad \|A\|_{\mathfrak{E}}^2 = \sum_{p \in \mathcal{P}} \|A_p\|_{\text{HS}}^2. \quad (1.1)$$

The raw innovations $\xi_{[p]}$ are distinct. A target may later be a raw source, but a later control still reads that raw source rather than the shifted state. Thus (1.1) is affine and one-shot.

Fix the side-16 A1 symbol, the positive A7 floor, exact continuum torus integration, and a common real-even covariance-matched scalar multiplier $|m_\Lambda(k)| \leq 1$ identical on all real components. Then there is one existential number $\delta_* > 0$, independent of

$$|\mathcal{P}|, \quad \Lambda < \infty, \quad m_\Lambda, \quad \text{and every factor-three path length,} \quad (1.2)$$

such that for every $\|A\|_{\mathfrak{E}} < \delta_*$ and nonzero direction H ,

$$\boxed{D^2 \mathcal{A}_{J,\mathcal{P}}(A)[H, H] > \frac{13}{100} \sum_{p \in \mathcal{P}} \|H_p\|_{\text{HS}}^2 > \frac{1}{10} \sum_{p \in \mathcal{P}} \|H_p\|_{\text{HS}}^2.} \quad (1.3)$$

Here $\mathcal{A}_{J,\mathcal{P}}$ is the *global terminal* source action. The R-107 predictable-baseline telescope identifies its endpoint with the one terminal V_j^{ren} difference before source and sextic are added. Equation (1.3) does not identify an individual root's covariance-normal atom with its smaller P_{comp} owner.

Equation (6.3) gives an analytic positive radius in terms of finite derivative suprema, but it is not evaluated numerically. The theorem gives no deterministic or realised Gaussian past, shifted-state recursion, nonlinear or revisited feedback, finite-amplitude convexity, floor removal, infinite-cutoff endpoint, T-050 or A13 closure, Nelson theorem, interacting measure, phase, vacuum, morphology, PDE verdict, or Sector-A closure.

2. Global terminal action and the local-owner firewall

This distinction is load-bearing. R-107 writes, on a temporally faithful finite chart,

$$Z_k = S_\pi \xi + \sum_{\ell \leq k} S_\ell h_\ell, \quad Z_0 = S_\pi \xi, \quad (2.1)$$

$$\Delta_k V = V_J^{\text{ren}}(Z_k) - V_J^{\text{ren}}(Z_{k-1}), \quad P_k = \mathbb{E}[\Delta_k V \mid \mathcal{F}_{k-1}], \quad (2.2)$$

and proves the exact source-action representation

$$\mathcal{A}_J(h) = \sum_k \mathbb{E} P_k + \frac{3}{20} \mathbb{E} \|Z_K\|_6^6 + \frac{9}{20} \mathbb{E} \sum_k \|h_k\|^2. \quad (2.3)$$

Tower property and telescoping give

$$\sum_k \mathbb{E} P_k = \mathbb{E} \{V_J^{\text{ren}}(Z_K) - V_J^{\text{ren}}(Z_0)\}. \quad (2.4)$$

The A7 zero-control convention and R-146 relative anchor give

$$\mathbb{E} V_J^{\text{ren}}(Z_0) = 0. \quad (2.5)$$

R-151's scalar $\mathcal{R}$ is precisely the global finite-cutoff A7 endpoint

$$\frac{1}{2} \int \mathbb{E} \left\{ \sum_i V_i^* \mathcal{B}(W) V_i - \text{Tr}[\mathcal{B}(W) Q_0] \right\} dx, \quad (2.6)$$

which is the V_J^{ren} convention in (2.4)–(2.5). Hence a Hessian bound obtained by differentiating (2.6) is a bound on the global terminal part of (2.3), not a lower-bound transfer from one local owner.

R-141 writes the same complete action as the exhaustive global sum of $\mathbb{E} \mathcal{P}_{\text{comp}, \alpha}$ plus the identical Cameron–Martin and terminal sixth-power terms. Comparing that identity with (2.3), after subtracting those two common terms, gives

$$\mathcal{R}(A) = \sum_k \mathbb{E} P_k = \sum_\alpha \mathbb{E} \mathcal{P}_{\text{comp}, \alpha} \quad (2.7)$$

only after the complete global telescope and exhaustive owner sum. This global equality does not identify any individual P_k , E_{CN} , or $\mathcal{P}_{\text{comp}}$ atom.

For one exhaustive local packet, R-141 instead has

$$E_{\text{CN}} = P_{\text{comp}} + \frac{1}{2} V_{\text{future}}, \quad P_{\text{comp}} = \frac{1}{2} \|\Phi\|^2 - \frac{1}{2} \Theta. \quad (2.8)$$

Therefore $E_{\text{CN}} \geq P_{\text{comp}}$ has the wrong direction for lower-bounding that local action owner. The existing AUDIT-2026-07-30-A13-COVARIANCE-NORMAL-DOMINANCE-ACTION-DIRECTION remains binding. Equations (2.1)–(2.7), rather than the forbidden identification $E_{\text{CN}} = P_{\text{comp}}$, are the owner bridge used below.

3. Infinite-envelope control space and affine covariance jets

It is convenient to place every finite family in the Hilbert direct sum over all nonzero antipodal lattice classes,

$$\mathfrak{E} = \ell^2 \left((\text{HS}_p)_{[p] \neq 0} \right), \quad (3.1)$$

For $p = (2\pi/16)n$, $n \in \mathbb{Z}^3 \setminus \{0\}$, R-160's exact lattice floor gives

$$x > \frac{3}{20}|n|^2 =: c_0|n|^2. \quad (4.7)$$

The max-norm shell $\|n\|_\infty = m$ contains $24m^2 + 2$ points. The integral-test bounds for the zeta sums yield

$$\sum_{n \neq 0} |n|^{-4} < 52, \quad \sum_{n \neq 0} |n|^{-6} < 35, \quad \sum_{n \neq 0} |n|^{-8} < 32. \quad (4.8)$$

No phase cancellation is used. From (4.6), the squared coefficient weights in (3.4) obey

$$\begin{aligned} (2s_p s_{2p})^2 &\leq \frac{g^2}{4x^4}, \\ (2d_p d_{2p})^2 &\leq \frac{g^2}{x^2}, \\ (s_{2p} d_p + s_p d_{2p})^2 &\leq \frac{5g^2}{8x^3}. \end{aligned} \quad (4.9)$$

Combining (4.7)–(4.9) gives the exact cardinality-independent constants

$$a_C^2 = 8g^2 c_0^{-4}, \quad a_Q^2 = 52g^2 c_0^{-2}, \quad a_K^2 = \frac{175}{8}g^2 c_0^{-3}. \quad (4.10)$$

The Q jet is limiting: its squared coefficients decay only as $|p|^{-4}$, which is summable in dimension three.

For the diagonal quadratic jets one may take

$$b_C = g, \quad b_Q = \frac{g}{2}, \quad b_K = g. \quad (4.11)$$

Indeed $4gx/(1+16x^2) \leq g/2$ is equivalent to $(4x-1)^2 \geq 0$, while $s_{2p}d_{2p} \leq g/\sqrt{2} < g$; the nonstrict value $b_K = g$ closes the common constant without an unresolved supremum. Cauchy–Schwarz in the p sum and (4.10)–(4.11) give, uniformly in every finite family and at every spatial point,

$$\begin{aligned} \|C(A) - C_0\| &\leq a_C r + b_C r^2, \\ \|Q(A) - Q_0\| &\leq a_Q r + b_Q r^2, \\ \|K(A)\| &\leq a_K r + b_K r^2, \end{aligned} \quad r = \|A\|_{\mathfrak{E}}. \quad (4.12)$$

The same calculation bounds the first two controller derivatives, uniformly, and (3.5) kills all higher covariance jets. The uncontrolled value covariance is also uniform because

$$\|C_{\text{bg}}\| \leq g\{1 + 52c_0^{-2}\}. \quad (4.13)$$

The raw derivative covariance need not be bounded as the cutoff grows. It is never estimated.

5. Covariance-normal cancellation and the uniform Hessian modulus

For every deterministic finite-support A , equations (3.2) are linear in the finite raw Gaussian vector. Hence $(W_A, V_{A,1}, V_{A,2}, V_{A,3})$ is one jointly centered Gaussian endpoint. Multiplication by two is injective on the dual lattice, and ordering classes by increasing norm makes every raw $[p]$ source available before its $[2p]$ target. When $[2p]$ is reused as a later source, the law retains its raw innovation rather than substituting the already shifted target. Thus the endpoint is both temporally faithful for the R-107 telescope and Gaussian for the identity below.

Write

$$\bar{B}(C) = \mathbb{E}_{N(0,C)} \mathcal{B}(W), \quad \bar{T}(C) = \mathbb{E}_{N(0,C)} D^2 \mathcal{B}(W). \quad (5.1)$$

Componentwise Gaussian integration by parts gives

$$\mathbb{E}[(V_i)_m(V_i)_n\mathcal{B}_{mn}(W)] = Q_{i,mn}\bar{\mathcal{B}}_{mn}(C) + (K_i)_{am}(K_i)_{bn}\bar{T}_{abmn}(C). \quad (5.2)$$

The A7 counterterm uses the same regulator-matched primitive covariance $Q_0 = Q(0)$. Subtracting it before taking a norm gives the all-state identity

$$2\mathcal{R}(A) = \int \left\{ \text{Tr}[(Q(A) - Q_0)\bar{\mathcal{B}}(C(A))] + \sum_i K_i(A) \otimes K_i(A) : \bar{T}(C(A)) \right\} dx. \quad (5.3)$$

This instantiates R-159's Gaussian lemma for the whole growing fixed-law union; it is not an extrapolation of an origin formula. The primitive Q_0 cancels algebraically before any norm is taken. This is why the linearly divergent sharp-cube derivative covariance does not enter (4.12).

At the fixed positive floor, every derivative of the six-row coefficient $\mathcal{B}(w)$ has polynomial growth. For a bounded positive semidefinite covariance set, Gaussian moments are uniformly bounded, and the heat identity

$$D_C^j \mathbb{E}_{N(0,C)} F(W)[E_1, \dots, E_j] = \frac{1}{2^j} (E_1)_{a_1 b_1} \cdots (E_j)_{a_j b_j} \mathbb{E}_{N(0,C)} [\partial_{a_1 b_1} \cdots \partial_{a_j b_j} F(W)] \quad (5.4)$$

shows that the required derivatives of $\bar{\mathcal{B}}$ and $\bar{T}$ are bounded and continuous, including at singular C . In particular, $D^3 \bar{T}$ uses at most $D^8 \mathcal{B}$, whose Gaussian moments are uniformly finite on the bounded covariance set. No C^{-1} score formula is used.

To avoid any appeal to compactness of the infinite-dimensional controller ball, put

$$\mathcal{Z} = \text{Sym}_6 \times \text{Sym}_6 \times (\mathbb{R}^{6 \times 6})^3$$

and equip this finite-dimensional pointwise covariance-data space with the norm

$$\|(C, U, K)\|_{\mathcal{Z}}^2 = \|C\|_{\text{op}}^2 + \|U\|_{\text{op}}^2 + \sum_i \|K_i\|_{\text{HS}}^2. \quad (5.5)$$

Here and in (4.10)–(4.12), $\|K\|^2 = \sum_i \|K_i\|_{\text{HS}}^2$. Define

$$a = (a_C^2 + a_Q^2 + a_K^2)^{1/2}, \quad b = (b_C^2 + b_Q^2 + b_K^2)^{1/2} = \frac{3g}{2}, \quad (5.6)$$

$$\eta(r) = ar + br^2, \quad L(r) = a + 2br, \quad d(r) = 2br. \quad (5.7)$$

For

$$Y_A(x) = (C(A, x), Q(A, x) - Q_0(x), K(A, x)), \quad (5.8)$$

equations (4.10)–(4.13) give the pointwise operator bounds

$$\|Y_A - Y_0\| \leq \eta(r), \quad \|DY_A\| \leq L(r), \quad \|DY_A - DY_0\| \leq d(r), \quad \|D^2 Y\| \leq 2b. \quad (5.9)$$

They hold in L_x^∞ , not merely after spatial integration, and are uniform in the family, cutoff and multiplier.

Define the universal pointwise scalar

$$\phi(C, U, K) = \frac{1}{2} \left\{ \text{Tr}[U\bar{\mathcal{B}}(C)] + \sum_i K_i \otimes K_i : \bar{T}(C) \right\}. \quad (5.10)$$

For $r \leq 1$, set

$$M_0 = g(1 + 52c_0^{-2}), \quad M_C = M_0 + a_C + b_C, \quad M_U = a_Q + b_Q, \quad M_K = a_K + b_K, \quad (5.11)$$

and define the explicit compact convex data box

$$\mathcal{K} = \{(C, U, K) : C \succeq 0, \|C\|_{\text{op}} \leq M_C, \|U\|_{\text{op}} \leq M_U, \|K\| \leq M_K\}. \quad (5.12)$$

Equations (4.12)–(4.13) put every $Y_A(x)$ in $\mathcal{K}$. This is compactness of covariance data, not compactness of the ℓ^2 controller ball. The positive A7 floor and (5.4) show that the heat derivatives $D^j\phi$, $j \leq 3$, extend continuously to the compact convex positive-semidefinite data set $\mathcal{K}$. At a singular covariance this follows by writing $W = C^{1/2}Z$, approximating by $C + \epsilon I$, and using dominated convergence together with the paired-index heat identity; $D^3\bar{T}$ requires only the already bounded Gaussian moments of $D^8\mathcal{B}$. The mean-value theorem below is used only along line segments inside $\mathcal{K}$. Consequently the finite constants

$$M_j = \sup_{Y \in \mathcal{K}} \|D^j\phi(Y)\|_{\text{op}} < \infty, \quad j = 2, 3, \quad (5.13)$$

are independent of $\mathcal{P}$, Λ and m_Λ .

The pointwise chain rule is

$$D^2(\phi \circ Y)_A[H, G] = D^2\phi(Y_A)[DY_A H, DY_A G] + D\phi(Y_A)[D^2Y(H, G)]. \quad (5.14)$$

Subtracting its value at zero, applying the mean-value theorem on $\mathcal{K}$, and using (5.9) gives the explicit operator estimate

$$\|D^2\mathcal{R}(A) - D^2\mathcal{R}(0)\|_{\text{op}} \leq \Omega(r) := 4096 \left\{ M_3\eta(r)L(r)^2 + 4M_2brL(r) + 2bM_2\eta(r) \right\}. \quad (5.15)$$

The three terms respectively change $D^2\phi$, its two arguments, and $D\phi$; no mode count occurs. For $0 \leq r \leq 1$, set

$$L_* = 4096 \left\{ M_3(a+b)(a+2b)^2 + 4M_2b(a+2b) + 2bM_2(a+b) \right\}. \quad (5.16)$$

Then (5.15) implies the uniform operator bound

$$\boxed{\sup_{\substack{\mathcal{P} \text{ finite, } \Lambda < \infty, |m_\Lambda| \leq 1 \\ \|A\|_{\mathfrak{E}} \leq r}} \|D^2\mathcal{R}_{\mathcal{P}, \Lambda}(A) - D^2\mathcal{R}_{\mathcal{P}, \Lambda}(0)\|_{\text{op}} \leq L_* r.} \quad (5.17)$$

This is the missing family-cardinality-free modulus at fixed physical dimension three. It is not a dimension-uniform field-theory estimate.

6. Gap retention and the once-owned sextic

R-160's actual side-16 lattice theorem gives, for every finite family,

$$D^2\{\mathcal{A}_{\text{CM}} + \mathcal{R}\}(0)[H, H] > \frac{4}{25}\|H\|_{\mathfrak{E}}^2. \quad (6.1)$$

Its complete sixth-power Hessian was dropped only as one global positive semidefinite form; (6.1) does not borrow positivity from a rootwise split.

The headroom over the T-050 comparison value is

$$\frac{4}{25} - \frac{1}{10} = \frac{3}{50}. \quad (6.2)$$

The analytic choice

$$\delta_* := \min \left\{ 1, \frac{3}{100(1 + L_*)} \right\} > 0 \quad (6.3)$$

satisfies $L_*\delta_* < 3/100$. Then (3.6), (5.17), and (6.1) give

$$D^2\{\mathcal{A}_{\text{CM}} + \mathcal{R}\}(A)[H, H] > \left(\frac{4}{25} - \frac{3}{100}\right) \|H\|_{\mathfrak{E}}^2 = \frac{13}{100} \|H\|_{\mathfrak{E}}^2. \quad (6.4)$$

For the affine field direction z_H , the remaining terminal term is

$$D^2\mathcal{S}_6(A)[H, H] = \frac{9}{10} \int \mathbb{E}[|W_A|^4 |z_H|^2 + 4|W_A|^2 (W_A \cdot z_H)^2] dx \geq 0. \quad (6.5)$$

It is retained once, globally. Adding (6.5) to (6.4) proves (1.3).

The same radius is valid for every finite cutoff, but this is not cutoff removal. No infinite-cutoff action, convergence of measures, or limiting operator domain is constructed.

7. Failed shortcuts and reusable boundaries

7.1 Per-family continuity is not enough

The affine scalar family

$$\mathcal{F}_N(h) = \frac{2}{25} \|h\|_2^2 - \frac{(\sum_j h_j)^4}{4N} + \frac{(\sum_j h_j)^6}{6N^2} \quad (7.1)$$

has origin Hessian $4I/25$ and a positive-semidefinite sextic Hessian. Nevertheless, for

$$h_N = (N^{-1}, \dots, N^{-1}), \quad e_N = N^{-1/2}(1, \dots, 1), \quad (7.2)$$

one has $\|h_N\|_2 = N^{-1/2} \rightarrow 0$ but

$$D^2\mathcal{F}_N(h_N)[e_N, e_N] = \frac{4}{25} - 3 + \frac{5}{N} = -\frac{71N - 125}{25N} < 0 \quad (N \geq 2). \quad (7.3)$$

This is a logical coherent-mode fixture, not an A1 counterexample. It shows why R-160 plus qualitative continuity could not prove R-161. The square-summable A1 jets in Section 4 are the new load-bearing input.

7.2 The proof is three-dimensional

The limiting controlled Q -jet square behaves as $|p|^{-4}$, and the background value-covariance bound (4.13) uses the same lattice exponent. Both sums are finite in three dimensions but logarithmically critical in dimension four. R-161 is not a dimension-independent field-theory theorem.

7.3 Local covariance-normal dominance is still unusable

Equation (2.8) remains a hard firewall. R-161 bounds the global terminal scalar through (2.1)–(2.6); it supplies no separate P_{comp} lower bound at each reveal and no permission to discard future variance in a relative owner calculation.

8. Devil's-advocate audit

1. Objection: a hidden factor $|\mathcal{P}|$ remains in the pointwise covariance sums. **DISMISSED.**

The linear sums use Cauchy–Schwarz against the full-lattice square sums (4.8)–(4.10); the quadratic sums use $\sum_p \|A_p\| \|H_p\|$. Neither estimate counts modes.

2. **Objection: oscillatory phases may destroy compactness. DISMISSED.** The proof takes pointwise absolute operator norms and uses no phase compactness or cancellation. Exact torus integration occurs only after the uniform pointwise estimate.
3. **Objection: the raw derivative covariance diverges with cutoff. DISMISSED.** Q_0 cancels exactly in (5.3). Only $Q - Q_0$, whose first jet is square summable, enters the modulus.
4. **Objection: Gaussian score differentiation uses C^{-1} and fails at a zero multiplier. DISMISSED.** Equation (5.4) uses derivative expectations and remains continuous at singular covariance.
5. **Objection: the global endpoint was silently substituted for a local P_{comp} owner. DISMISSED WITH A FIREWALL.** Equations (2.1)–(2.6) pin the R-107 global telescope and A7 anchor. Equation (2.8) is retained explicitly and no local transfer is claimed.
6. **Objection: the dense sixth-power Hessian was assigned to the factor-three graph. DISMISSED.** It is not used in the graph or the modulus loss. Equation (6.5) is one complete global PSD form.
7. **Objection: source-target reuse creates cross-source quadratic jets. DISMISSED in the declared fixed-law chart.** Distinct raw innovations remain independent. The conclusion would fail for a later control that reads a shifted state.
8. **Objection: cutoff-uniform radius proves cutoff removal. UPHELD AS PROHIBITED.** A common finite-cutoff estimate does not construct the infinite endpoint, limiting operator, Gibbs measure, or Nelson bound.
9. **Objection: a local affine theorem closes T-050. UPHELD AS PROHIBITED.** Realised past, recursive and revisit feedback, nonlinear predictable laws, the progressive owner decomposition, and the complete conditional Hessian remain open.
10. **Objection: the theorem selects BCC or any other phase/PDE. UPHELD AS PROHIBITED.** R-161 is phase-neutral source-Hessian mathematics and contains no candidate-state comparison.

External review is invited for the R-107-to-R-151 global-owner bridge, the antipodal and six-real normalization, the exact symbol envelope, max-shell counts, the limiting Q -jet exponent, source-target-reuse independence, Gaussian heat differentiation at singular covariance, operator-norm chain rule, global sextic ownership, and every local-versus-global scope boundary.

9. Reproduction and Result footer

Dependencies: A1, A7; R-107, R-141, R-145, R-146, R-151, R-155, R-159, R-160; AUDIT-2026-07-30-A13-COVARIANCE-NORMAL-DOMINANCE-ACTION-DIRECTION.

Reproduction command:

```
& E:\Dev\TECT.venv\Scripts\python.exe '
codes/foundations/a13_classii_summable_jet_growing_affine_root_union_
uniform_neighborhood_gap_boundary_verify.py
```

Result ID: A13-CLASSII-SUMMABLE-JET-GROWING-AFFINE-ROOT-UNION-
UNIFORM-NEIGHBORHOOD-GAP-BOUNDARY; Ledger ID: R-161.

Precise statement: one existential $l_2(\text{HS})$ radius, uniform over every finite family cardinality, finite cutoff, admitted regulator and factor-three path, retains the global terminal action Hessian gap $>13/100 > 1/10$.

Scope: side-16, fixed A1 symbol and A7 floor, centered no-past common-even
covariance-matched exact-torus one-shot fixed-law affine p:2p chart.
Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.
Expected output: primary, non-importing independent, and integrated PASS;
deterministic PDF and full release PASS.
Falsification gate: any owner bridge, symbol, shell, jet, Gaussian identity,
gap, hash, PDF, public-surface, exploration, or release assertion fails.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: no numerical radius, local E_CN=Pcomp transfer,
realised past, shifted-state recursion, nonlinear/revisit, removal, T-050,
A13, Nelson, measure, phase/PDE, or Sector-A closure.
Proof complete: false; T-050 closed: false; Sector A closed: false.
Next required action: instantiate a realised-past or recursive/revisit
production law and certify its complete conditional Hessian or counterexample.